Status	Finished
Started	Tuesday, 6 October 2026, 11:40 AM
Completed	Tuesday, 6 October 2026, 11:43 AM
Duration	3 mins 2 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 2.00 out of 2.00

Write a function that returns the value of a+aa+aaa+aaaa with a given digit as the value of a.

Suppose the following input is supplied to the program:

9

Then, the output should be:

9+99+999+9999=11106

Sample Input Format:

9

Sample Output format:

11106

For example:

Test	Result
print(Summation(8))	9872

Answer: (penalty regime: 0 %)

Reset answer

```
def Summation(n):
    # Convert the input to a string format to easily replicate characters
    digit_str = str(n)

    # Generate each term in the pattern by multiplying the string and casting to integer
    term1 = int(digit_str)
    term2 = int(digit_str * 2)
    term3 = int(digit_str * 3)
    term4 = int(digit_str * 4)

    # Return the total sum of all four terms
    return term1 + term2 + term3 + term4

```

	Test	Expected	Got	
✓	print(Summation(8))	9872	9872	✓
✓	print(Summation(10))	10203040	10203040	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **2**

Correct

Mark 2.00 out of 2.00

A number is considered to be ugly if its only prime factors are 2, 3 or 5.

[1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, ...] is the sequence of ugly numbers.

Task:

complete the function which takes a number n as input and checks if it's an ugly number.

return ugly if it is ugly, else return not ugly

Hint:

An ugly number U can be expressed as: U = 2^a * 3^b * 5^c, where a, b and c are nonnegative integers.

For example:

Test	Result
`print(checkUgly(6))`	`ugly`
`print(checkUgly(21))`	`not ugly`

Answer: (penalty regime: 0 %)

Reset answer

```
def checkUgly(n):
    # Ugly numbers must be positive integers
    if n <= 0:
        return "not ugly"

    # Divide out all factors of 2, 3, and 5
    for factor in (2, 3, 5):
        while n % factor == 0:
            n //= factor

    # If the remaining number is 1, it's an ugly number
    return "ugly" if n == 1 else "not ugly"


```

	Test	Expected	Got	
✓	`print(checkUgly(6))`	`ugly`	`ugly`	✓
✓	`print(checkUgly(21))`	`not ugly`	`not ugly`	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **3**

Correct

Mark 2.00 out of 2.00

complete function to implement coin change making problem i.e. finding the minimum

number of coins of certain denominations that add up to given amount of money.

The only available coins are of values 1, 2, 3, 4

Input Format:

Integer input from stdin.

Output Format:

return the minimum number of coins required to meet the given target.

Example Input:

16

Output:

4

Explanation:

We need only 4 coins of value 4 each

Example Input:

25

Output:

7

Explanation:

We need 6 coins of 4 value, and 1 coin of 1 value

Answer: (penalty regime: 0 %)

Reset answer

```
def coinChange(n):
    # If the target amount is 0 or negative, 0 coins are needed
    if n <= 0:
        return 0

    # Use integer division to get the maximum number of 4-value coins
    coins_of_four = n // 4
    remainder = n % 4

    # If there is a remainder, it can always be fulfilled by exactly 1 coin
    # since our denominations include all remaining possibilities (1, 2, or 3)
    if remainder > 0:
        return coins_of_four + 1

    return coins_of_four
```

	Test	Expected	Got	
✓	print(coinChange(16))	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **4**

Correct

Mark 2.00 out of 2.00

An e-commerce company plans to give their customers a special discount for Christmas.

They are planning to offer a flat discount. The discount value is calculated as the sum of all

the prime digits in the total bill amount.

Write an algorithm to find the discount value for the given total bill amount.

Constraints

1 <= orderValue< 10e100000

Input

The input consists of an integer orderValue, representing the total bill amount.

Output

Print an integer representing the discount value for the given total bill amount.

Example Input

578

Output

12

For example:

Test	Result
print(christmasDiscount(578))	12

Answer: (penalty regime: 0 %)

Reset answer

```
def christmasDiscount(n):
    # Convert input to string to handle extremely large numbers efficiently
    bill_str = str(n)

    # Map each prime digit to its integer value for O(1) sum lookup
    prime_values = {'2': 2, '3': 3, '5': 5, '7': 7}

    discount = 0
    for digit in bill_str:
        if digit in prime_values:
            discount += prime_values[digit]

    return discount
```

	Test	Expected	Got	
✓	print(christmasDiscount(578))	12	12	✓
✓	print(christmasDiscount(57))	12	12	✓
✓	print(christmasDiscount(222))	6	6	✓
✓	print(christmasDiscount(77777))	35	35	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **5**

Correct

Mark 2.00 out of 2.00

A strobogrammatic number is a number that looks the same when rotated 180 degrees (looked at upside down).

Write a program to determine if a number is strobogrammatic. The number is represented as a string.

Example 1:

Input:

69

Output:

true

Example 2:

Input:

88

Output:

true

Example 3:

Input:

962

Output:

false

Example 4:

Input:

1

Output:

true

For example:

Test	Result
`print(Strobogrammatic(69))`	`True`
`print(Strobogrammatic(962))`	`False`

Answer: (penalty regime: 0 %)

Reset answer

```
def Strobogrammatic(n):
    # Convert input to string to handle both integer and string inputs safely
    n = str(n)

    # Define the valid strobogrammatic pairs
    lookup = {'0': '0', '1': '1', '6': '9', '8': '8', '9': '6'}

    # Initialize two pointers at both ends of the string
    left, right = 0, len(n) - 1

    # Loop towards the middle of the string
    while left <= right:
        # If the character isn't a valid flippable digit, it's not strobogrammatic
        if n[left] not in lookup:
            return False

        # Check if the flipped left character matches the right character
        if lookup[n[left]] != n[right]:
            return False

```

```
        # Move pointers inward
        left += 1
        right -= 1

    return True

```

	Test	Expected	Got	
✓	print(Strobogrammatic(69))	True	True	✓
✓	print(Strobogrammatic(88))	True	True	✓
✓	print(Strobogrammatic(962))	False	False	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.